

Proactive Memory for Event-Driven LLM Agents

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Abstract

Event-driven LLM agents must decide whether incoming observations warrant assistance, often using evidence spread across earlier events. Retaining related history is not enough: the agent must assess its support and current applicability without treating recurrence alone as sufficient reason to intervene. We propose Proactive Evidence Memory (ProEviMem), combining bounded recent events, source-checked daily trajectories, and replay-assessed temporal patterns. The framework checks daily facts against their sources and assesses patterns against prior observations, then selects applicable evidence for the agent’s service-opportunity judgment. Across ProactiveStream’s three application scopes, ProEviMem improves Intervention F1 by 4.3–11.5 percentage points over the strongest baseline in each scope, among comparators including recursive-summary and temporal-graph methods. Hierarchy and validation ablations support the contribution of evidence organization and historical support checks to opportunity detection.

1 Introduction

An activity record, a home-sensor update, or a repository change may warrant assistance before a user asks for it. Yet a similar observation may require no action: its significance depends on prior events and whether the opportunity has already been addressed. Event-driven LLM agents must therefore judge not only what an event means, but whether it warrants intervention now.

Agent memory systems retain observations, consolidate experience, and retrieve related information (Park et al., 2023; Packer et al., 2023; Chhikara et al., 2025). Recent work also studies history-based intervention judgment and proactive memory use (Lu et al., 2025; Xie et al., 2026; Li et al., 2026; Wu et al., 2026). These capabilities make historical information available, but its use for intervention requires three distinct judgments: whether

an abstraction reflects its source events, whether a proposed temporal pattern has recurring support, and whether it applies now. A similar past event alone establishes neither recurrence nor a present service need. The central question is thus *how to turn retained history into evidence for a timely intervention*, rather than simply how much history to retain.

We propose Proactive Evidence Memory (ProEviMem; Figure 1) to separate evidence construction from intervention judgment. Bounded Event Memory preserves recent observations; source-checked Daily Trajectory Memory retains evidence beyond that window; Temporal Pattern Memory captures recurring conditions assessed through historical replay. At decision time, the agent combines event and daily evidence with currently applicable patterns. Either the current event or historical evidence may support a concrete, timely, low-risk service; recurrence is neither required nor an instruction to intervene. Episode control then checks whether the proposed opportunity has already been covered.

We evaluate ProEviMem on *ProactiveStream*: 4,539 events from 17 isolated streams across Activity Support, Ambient Living, and Repository Assistance. Under causal replay, it improves Intervention F1 by 4.3–11.5 percentage points over the strongest baseline in each scope, among ProactiveAgent, Mem0, Cross-week Summary Memory (CSM), and Graphiti. Hierarchy and validation ablations examine the contribution of evidence organization and checks beyond history retention alone.

Our contributions are:

- **Method.** An evidence lifecycle separating source fidelity, historical support, and current applicability from intervention judgment.
- **Evaluation platform.** ProactiveStream supports causal, entity-isolated evaluation with episode-aware reference opportunities across three service scopes.

- **Empirical analysis.** Memory-interface comparisons, targeted ablations, and behavioral diagnostics characterize the contributions and limits of the proposed lifecycle.

2 Problem Formulation

Let u denote an isolated entity with a time-ordered public event stream

$$E^u = (e_1^u, \dots, e_{T_u}^u), \quad e_t^u = (\tau_t^u, x_t^u), \quad (1)$$

where T_u is the number of events for entity u , τ_t^u is the event time, and x_t^u contains the scope-specific public fields observable at that time. We omit the entity superscript u when it is unambiguous. Before processing e_t^u , an agent has a causal state Z_t^u constructed only from events and finalized outputs preceding e_t^u in the fixed stream order (including a fixed order for timestamp ties). It produces

$$(\hat{y}_t, \hat{m}_t) = \pi(e_t^u, Z_t^u), \quad \hat{y}_t \in \{0, 1\}, \quad (2)$$

where $\hat{y}_t = 1$ denotes INTERVENE, $\hat{y}_t = 0$ denotes ABSTAIN, and $\hat{m}_t$ is an optional evidence-grounded service message. Here π denotes a general inference policy over the current event and causal state; $\hat{m}_t$ may be empty under ABSTAIN. The event is committed only after the output is finalized:

$$Z_{t+1}^u = U(Z_t^u, e_t^u, \hat{y}_t, \hat{m}_t). \quad (3)$$

No state is shared across entities, and neither future events nor evaluator-private information is available to π .

The objective is to identify a timely, actionable *service opportunity* supported by the current event or prior same-entity observations, not unusual behavior alone. Let $y_t \in \{0, 1\}$ denote the frozen, reviewed label for e_t , defined independently of the agent’s state representation (Section 4.2). Labels and annotation metadata remain evaluator-private; intervention and abstention events share the same scope-specific public schema.

3 Proactive Evidence Memory (ProEviMem)

ProEviMem organizes recent public events, fact-validated daily trajectories, and replay-assessed temporal patterns into three evidence levels (Figure 1). Construction checks source fidelity and historical support; retrieval selects applicable patterns; the decision LLM judges present service value.

3.1 Memory Construction and Update

Bounded Event Memory. For event e_t , Event Memory retains prior events without semantic rewriting, intersecting a temporal horizon Δ with capacity K :

$$S_t = \text{Tail}_{K-1}((e_i)_{i < t, 0 \leq \tau_t - \tau_i \leq \Delta}). \quad (4)$$

We use $\Delta = 72$ hours and $K = 100$, including the separately supplied current event. An independent buffer retains all committed events of the open local day for daily construction, unaffected by this truncation.

Fact-validated Daily Trajectory Memory. After a local day closes, a construction LLM summarizes its buffered events as trajectory segments with intervals, typed facts, and event citations. For segment d , normalized checkable facts $F(d)$ and cited events $\mathcal{E}(d)$ must satisfy

$$F(d) \subseteq \bigcup_{e \in \mathcal{E}(d)} F(e), \quad \mathcal{E}(d) \subseteq E_{\text{completed day}}^u. \quad (5)$$

The validator checks entity, interval, citations, and canonicalized scalar values at compatible field paths. Unsupported facts and unresolved citations are rejected; free text cannot substitute for typed evidence. Accepted daily trajectories form W_t , preserving evidence beyond Event Memory. Appendix E details field handling and buffering.

Replay-assessed Temporal Pattern Memory. A weekly cycle is a non-overlapping block of seven completed daily records for one entity, not necessarily a calendar week. An LLM proposes a trigger g , expected pattern p , time window, optional weekday restrictions, and source daily identifiers. Replay checks these conditions against validated daily facts: an observable triggered day is fulfilled when p occurs at or after the trigger within the window and violated otherwise. Missing observations are censored, inapplicable days are excluded, and each date contributes at most once. For fulfilled and violated day counts n_f, n_v ,

$$\text{conf}_{\text{replay}} = \frac{n_f}{n_f + n_v}, \quad \text{lift} = \frac{\text{conf}_{\text{replay}}}{\text{base}(p)}. \quad (6)$$

The base rate is the frequency of p over observable days under the same window/weekday restrictions without conditioning on g . Zero denominators yield zero confidence or lift, respectively.

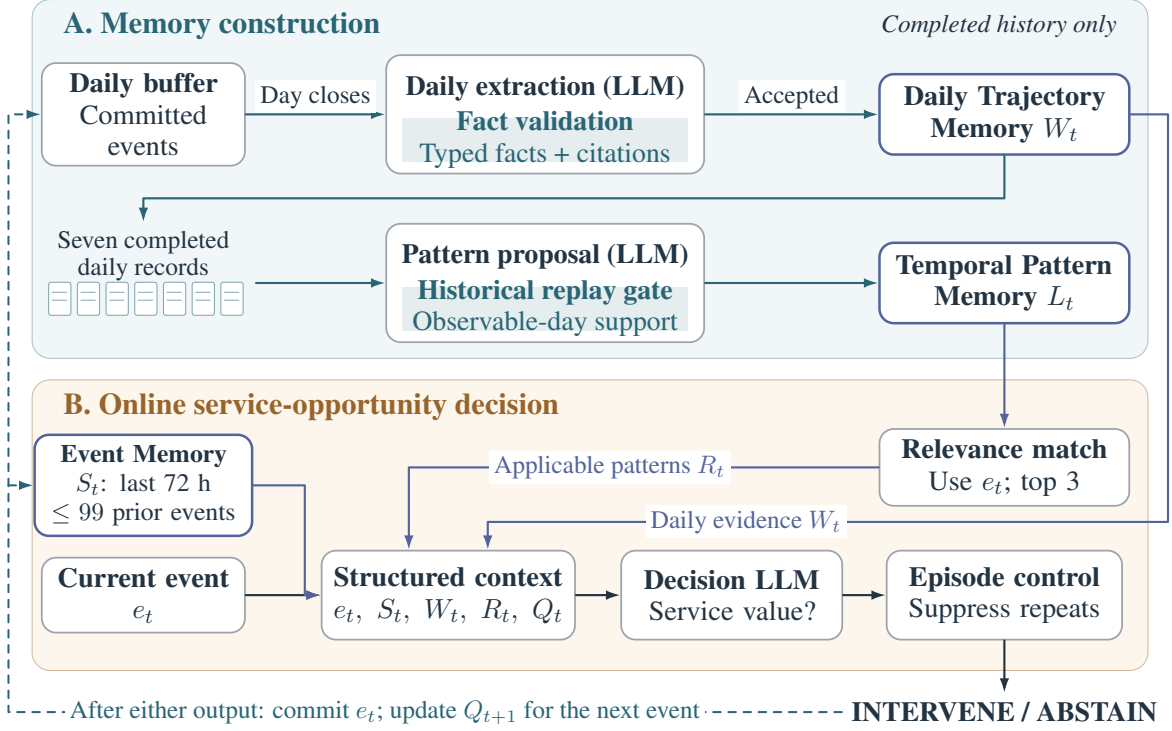


Figure 1: ProEviMem separates memory construction (A) from online decision-making (B). Daily extraction is checked against cited events; temporal proposals are assessed by replay over completed daily records. Only lifecycle-eligible, currently relevant patterns enter the structured context. The decision LLM assesses service value, and the shared controller can suppress repeated interventions but cannot create them. Dashed feedback commits the current event after the output is finalized, whether INTERVENE or ABSTAIN; Q_t records prior episode coverage.

A well-formed proposal becomes *Active* with at least three supporting days, confidence ≥ 0.5 , and lift ≥ 1.0 ; otherwise it remains decision-invisible *Candidate*. Malformed proposals are rejected. Consolidation resets the online daily view, retaining records for audit. Later observations update pattern confidence and lifecycle: repeated violations can lower retrieval priority or suspend eligibility. Appendix E.3 specifies these transitions. Historical support does not establish future validity or require intervention.

3.2 Memory Retrieval and Selection

Event Memory supplies the bounded sequence and W_t supplies all completed daily trajectories in the current cycle. Pattern retrieval additionally checks entity-local trigger context, weekdays, and temporal applicability with a 30-minute lead/grace interval. Current and previously recorded causal trigger occurrences are eligible; fulfilled and unobservable (censored) occurrences are not. For pattern store L_t^u , lifecycle eligibility Eligible_t , and occurrence

state s_t ,

$$R_t = \text{Top3} \{ \ell \in L_t^u : \text{Eligible}_t(\ell) \wedge \text{Match}_t(\ell), s_t(\ell) \notin \{\text{fulfilled, censored}\} \}. \quad (7)$$

Ranking prioritizes occurrence state, lifecycle, confidence, and temporal distance (Appendix E.4). Selection uses no future observations: historical support and present applicability are separate requirements.

3.3 Proactive Decision and Episode Control

The decision context separates current observations, daily evidence, and selected patterns:

$$C_t = (e_t, S_t, W_t, R_t, Q_t), \quad (8)$$

where Q_t records observable episode state and prior coverage. The LLM produces $(\tilde{y}_t, \tilde{m}_t) = \pi_\theta(C_t)$, proposing intervention when the current event *or* prior evidence supports concrete, timely, feasible, low-risk assistance, and abstention otherwise. A deviation is neither required nor sufficient. The shared Episode Controller may suppress, but never create, an intervention. It identifies opportunities

by observable state, service intent, and pattern evidence; changed state, stronger evidence, changed service, or a new local day permits reconsideration, not automatic intervention. All methods use the same controller policy, applied to their own observable state and available evidence; Appendix E.5 specifies identity construction.

Inference prepares prior memory, stages e_t , selects evidence, obtains and finalizes the decision, and only then commits e_t . Maintenance incorporating e_t can affect only later decisions. Reference labels enter neither construction nor retrieval.

4 ProactiveStream Benchmark

ProactiveStream instantiates the causal INTERVENE/ABSTAIN task across three scopes, with scope-specific observations and services under a common opportunity-labeling principle. Table 1 reports all 4,539 events, 17 isolated streams, and 729 reference interventions.

4.1 Scopes and Public Events

Activity Support (AS). HeartSteps mobile activity contexts (Klasnja et al., 2019) provide nine complete participant streams. Events expose contemporaneous availability, activity, step summaries, location, weather, and local slot, excluding randomized treatment assignments and future outcomes. Labels concern optional activity support, not the original treatment policy or health effects.

Ambient Living (AL). Four CASAS homes (Cook et al., 2013) each contribute 14 consecutive days. Within-day ten-minute blocks form 28 regular 50-minute segments and one 40-minute tail per day: 406 events per home. Segments become available at interval closure and describe only within-interval sensor activity, without historical baselines. Services are limited to gentle check-ins, excluding disease, falls, danger, and unsupported personal inferences.

Repository Assistance (RA). We organize repository activity records into chronologically ordered event streams. Each repository defines an independent stream containing commit pushes, issue updates, pull-request actions, and reviews. We preserve GitHub’s native event granularity, with each event corresponding to one intervention decision. Memory construction and intervention decisions are performed separately for each repository, without transferring observations or memory across

repositories. Intervention history likewise remains repository-local, even when contributors overlap. Services include low-risk triage, review, and maintenance. Native PushEvent denotes repository activity, not an intervention.

Across scopes, both label classes use the same scope-specific public event schema. Private reference labels, candidate flags, episode annotations, trajectory summaries, memory-level markers, and scoring masks are absent. Appendix F gives detailed fields and preprocessing.

4.2 Reference Labels and Causal Replay

Annotators view e_t and complete strictly prior same-entity history $\mathcal{H}_{<t}$, independently of a tested agent’s memory representation. Let A denote a concrete, timely, feasible, low-risk opportunity visible in e_t , B one made visible by prior evidence, and N_t^{ref} reference-episode novelty:

$$y_t = 1 \iff (A(e_t) = 1 \vee B(e_t, \mathcal{H}_{<t}) = 1) \wedge N_t^{\text{ref}} = 1. \quad (9)$$

These are annotation predicates, not additional model gates. The first reliable actionable window is INTERVENE; covered windows are ABSTAIN unless evidence strengthens, service changes, or the episode reopens. Reference coverage follows the reviewed opportunity sequence, not model predictions.

LLM-assisted screening prioritizes review but never assigns final labels automatically. Three team members independently annotate without evaluated-system decisions, then cross-review and adjudicate every event. For the main comparisons, we use protocol-fixed evaluation, not an independent development/test split: aggregation precedes row construction, and versioned labels, prompts, memory bounds, and validation settings are frozen before those runs. Labels represent reviewed service opportunities, not recipient acceptance.

Each stream starts with empty method state and is replayed chronologically. Only current public fields and strictly prior same-stream state are accessible. Decisions precede memory commits; reference labels are joined only after inference. Early events before consolidated memory becomes available remain scored.

Scope	Primary source	Entity	Streams	Input events	Temporal coverage	Reference interventions
AS	HeartSteps	participant	9	1,456	367 observed participant-days	365
AL	CASAS	home	4	1,624	14 consecutive days/home	190
RA	GH Archive	repository	4	1,459	longitudinal repository histories	174
Total	three scopes	isolated stream	17	4,539	three longitudinal settings	729

Table 1: Composition and provenance of ProactiveStream. AS: Activity Support; AL: Ambient Living; RA: Repository Assistance. Each stream is an independent memory and replay unit. Reference interventions denote team-reviewed service opportunities rather than observed recipient acceptance or downstream benefit.

5 Experimental Evaluation

5.1 Experimental Settings

We evaluate method comparisons (**RQ1**), memory-level contributions (**RQ2**), evidence-validation mechanisms (**RQ3**), and model sensitivity and errors (**RQ4**).

Comparators. *ProactiveAgent* uses the released evaluation prompt and proposal/null contract without its fine-tuned checkpoint, retaining causal event–decision history capped at 1,000 events (Lu et al., 2025; THUNLP, 2026). *Mem0* uses open-source extraction, consolidation, and same-entity retrieval (Chhikara et al., 2025; Mem0, 2026). Both share the task semantics and final Episode Controller, but retain their native prompt/memory interfaces; budgets are not equalized. *Cross-week Summary Memory (CSM)* is our plain-summary baseline: it recursively combines completed daily summaries with an earlier cross-week summary, without typed fact validation, replay gating, or explicit temporal-rule matching. *Graphiti* adapts an open-source temporal graph to store event-derived relations and retrieve same-stream facts (Zep, 2026). Both augment bounded Event Memory and reuse the task policy and controller, with memory-specific interface guidance rather than identical prompt text. *Event-only*, *Event+Daily*, and *ProEviMem* share the decision prompt, Event Memory bound, schema, and controller, differing only in memory levels. For AS/AL, *Flat Retrieval Memory* adds three similar older same-entity events to Event-only using fixed dense embeddings and cosine similarity, without consolidation or replay. Section 5.4 defines the mechanism ablations; Appendix G lists the recorded interface and retrieval settings.

Model and protocol. The primary backbone is gpt-5.6-luna; the original ProEviMem decision and construction runs use the official OpenAI API, with temperature zero and thinking disabled. Decision and construction are separate roles with separate output budgets. Terra/Sol comparisons replace both roles together (Appendix A). Input manifests, prompts, schemas, thresholds, and configurations are versioned.

Metrics. Across all scopes, INTERVENE includes both immediate and scheduled service decisions; ABSTAIN denotes no intervention decision at the current event. We aggregate TP/FP/FN/TN within each scope, then compute Intervention Rate, Precision, Opportunity Recall, Intervention F1, False Intervention Rate (FIR), and Balanced Accuracy (BA). Mean rows equally average the reported scope metrics, not pooled counts or entity scores. FIR is $FP/(FP+TN)$, not $1 - \text{Precision}$; BA averages recall and specificity. Higher precision, recall, F1, and BA and lower FIR are preferred. Intervention Rate describes decision frequency and has no universal preferred direction. Events are scored after shared episode control; a separate RA diagnostic also scores the recorded proposals before suppression.

5.2 Main Results

ProEviMem leads in Intervention F1 and Balanced Accuracy in every scope (Table 2, RQ1). Against the strongest baseline per scope—Mem0 on AS and CSM on AL/RA—its F1 gains are 4.3, 6.8, and 11.5 points, respectively.

Coverage and selectivity. The gains do not reflect uniformly more intervention. On AS, ProEviMem reduces FIR against all four baselines, with less recall than ProactiveAgent and Mem0. On AL, it improves both precision and recall over all base-

Scope	Method	Intervention Rate	Precision $\uparrow$	Opportunity Recall $\uparrow$	Intervention F1 $\uparrow$	False Intervention Rate $\downarrow$	Balanced Accuracy $\uparrow$
AS	Graphiti	55.1	29.3	64.4	40.3	52.0	56.2
	CSM	58.9	30.8	72.3	43.2	54.4	59.0
	ProactiveAgent	69.0	30.0	82.5	43.9	64.5	59.0
	Mem0	85.2	28.8	97.8	44.5	81.0	58.4
	ProEviMem	53.8	35.8	76.7	48.8	46.1	65.3
AL	Mem0	8.5	23.9	17.4	20.1	7.3	55.0
	ProactiveAgent	9.2	34.2	26.8	30.1	6.8	60.0
	Graphiti	17.9	33.4	51.1	40.4	13.5	68.8
	CSM	17.1	43.7	63.7	51.8	10.9	76.4
	ProEviMem	17.7	48.6	73.7	58.6	10.3	81.7
RA	ProactiveAgent	53.1	17.5	78.2	28.7	49.7	64.2
	Mem0	7.7	39.8	25.9	31.4	5.3	60.3
	Graphiti	28.1	26.8	63.2	37.7	23.3	69.9
	CSM	39.5	28.6	94.8	43.9	32.1	81.4
	ProEviMem	26.7	40.1	89.7	55.4	18.1	85.8
Mean	Mem0	33.8	30.8	47.0	32.0	31.2	57.9
	ProactiveAgent	43.8	27.2	62.5	34.2	40.3	61.1
	Graphiti	33.7	29.9	59.6	39.5	29.6	65.0
	CSM	38.5	34.4	76.9	46.3	32.4	72.3
	ProEviMem	32.7	41.5	80.0	54.3	24.8	77.6

Table 2: Main comparison (%). AS: Activity Support; AL: Ambient Living; RA: Repository Assistance. Mean equally averages scope metrics, not pooled counts. Baselines are ordered by F1 within each group; ProEviMem is last. Bold marks the best directional score, including the lowest FIR. Intervention Rate is not ranked.

lines; FIR increases relative to ProactiveAgent and Mem0 but decreases relative to CSM and Graphiti. On RA, it recovers more opportunities with lower FIR than ProactiveAgent, while Mem0’s infrequent interventions leave many opportunities uncovered. Intervention Rate is thus an operating point, not a performance target. ProactiveAgent intervenes on 9.2% of AL events versus 53.1% of RA events despite similar reference-positive prevalences (11.7% versus 11.9%). Appendix G.1 decomposes this difference without attributing it solely to the LLM.

Summary and graph memory. CSM exceeds Event-only by 4.9, 11.8, and 15.2 F1 points on AS, AL, and RA, supporting plain-summary retention. ProEviMem remains ahead of CSM by 5.5, 6.8, and 11.5 points. On RA, CSM recovers nine more reference opportunities (165 versus 156 TP), but incurs 179 more FP (412 versus 233); ProEviMem’s advantage is selectivity rather than maximum coverage. Graphiti reaches 40.3%, 40.4%, and 37.7% F1: temporal relation retrieval does not match ProEviMem under this adapter. These memory-interface comparisons do not isolate replay’s contribution; Section 5.4 tests it directly.

Retrieval and model controls. Flat Retrieval Memory attains 41.3% F1 on AS and 48.2% on

Method	Metric	AS	AL	RA	Mean
Event-only	F1 $\uparrow$	38.3	40.0	28.7	35.7
	FIR $\downarrow$	51.3	12.4	18.9	27.5
Event+Daily	F1 $\uparrow$	38.9	45.1	48.5	44.2
	FIR $\downarrow$	51.3	12.4	19.1	27.6
ProEviMem	F1 $\uparrow$	48.8	58.6	55.4	54.3
	FIR $\downarrow$	46.1	10.3	18.1	24.8

Table 3: Hierarchy ablation (%): Intervention F1 and False Intervention Rate (FIR). Bold marks the best score. Mean averages displayed scope scores; Appendix H adds net TP increments.

AL, below ProEviMem by 7.4 and 10.4 points despite similar intervention rates. ProEviMem has higher precision and recall in both scopes. Jointly replacing decision and construction LLMs from Luna to Terra to Sol increases F1 and decreases FIR in every scope (RQ4; Appendix A).

5.3 Memory-Hierarchy Ablation

Table 3 holds the Luna backbone, Event Memory, decision prompt, schema, and controller fixed. Daily evidence gives its largest F1 increment on RA (19.7 points). Temporal Pattern Memory adds 9.9, 13.5, and 7.0 points on AS, AL, and RA, respectively.

Daily evidence increases TP by 4/15/64 on

AS/AL/RA, with unchanged FP on AS/AL and two additional FP on RA; hence FIR stays flat or slightly rises despite higher F1. Adding patterns then gains 57/33/22 TP while reducing FP by 57/30/12, improving both coverage and selectivity. These net aggregate changes support complementary roles within the hierarchy (RQ2), not individual-event attribution or a replay-only effect.

5.4 Mechanism Ablations and Controller Analysis

The validation ablations hold Luna, Event Memory, decision prompts, context capacity, and episode control fixed (Table 4). *w/o Replay Gate* admits well-formed, source-linked, online-reconstructable Candidates to relevance matching without Equation 6’s admission thresholds; matching and decision judgment remain enabled. *w/o Daily Validation* bypasses the typed-fact check for schema-valid, same-entity daily extractions, retaining weekly construction and replay. The RA-only *Before suppression* diagnostic scores recorded model proposals from the same run, without changing its memory or later decisions.

Evidence checks. Removing replay gating increases Intervention Rate and reduces precision, recall, and F1 in all scopes. Removing daily validation also lowers precision/F1, while recall is unchanged on AS/RA and slightly higher on AL. On RA, retaining replay changes 145 TP/284 FP to 156 TP/233 FP; daily validation retains 156 TP while reducing FP from 276 to 233. The larger F1 loss without replay appears in every scope, supporting both evidence checks beyond store consistency (RQ3).

Episode control. On the recorded RA sequence, episode suppression changes 173 TP/299 FP to 156 TP/233 FP: 66 fewer FP but 17 fewer TP. Precision and F1 rise by 3.5 and 1.9 points, while recall falls from 99.4% to 89.7% and BA from 88.1% to 85.8%. For these recorded proposals, suppression trades coverage for precision/F1 and can withhold reference-positive interventions.

5.5 Evidence Availability and Error Analysis

Construction and availability. Across 13 AS/AL entities, 423 daily trajectories contain 1,279 source-supported segments; logs record 311 rejected checks and no retained cross-entity evidence. The pattern store has 42 Active and 12 Candidate proposals with replay-consistent records

(Appendix B). Median times to first daily evidence, Active pattern, and relevant pattern are 0.80, 6.79, and 10.46 days among the 13, 11, and eight entities reaching each stage, respectively. Activation therefore does not ensure current relevance.

A logged AS timeline links source events, daily extraction, pattern admission, and a matched context ending in ABSTAIN (Appendix C). It illustrates evidence flow and service-value judgment (RQ4), not counterfactual memory necessity.

Human-review diagnostic. A separate three-reviewer team, blinded to model outputs, reviews one contiguous 150-event segment per scope with its complete preceding history. Agreement with earlier diagnostic references is 88.7%, 91.3%, and 85.3% in AS, AL, and RA. Holding each method’s decisions fixed, reviewed-label F1 is 42.5%/55.6%/47.4% for ProEviMem versus 41.0%/37.2%/46.5% for CSM (Appendix D). The AL/RA ordering is preserved, but the small AS difference reverses: most labels agree, yet opportunity boundaries affect this sampled comparison.

Episode diagnostic. A provisional paired analysis of CSM and ProEviMem separates first assistance, valuable follow-up, and redundancy on the same 450 targets (Appendix D.1). Under the shared proposed references, ProEviMem has higher supported-intervention precision across scopes, but CSM covers more RA opportunities; this diagnoses coverage–selectivity trade-offs rather than uniform superiority.

6 Related Work

Agent memory and long-horizon retrieval. Generative Agents, MemoryBank, and MemGPT organize observations and experience through reflection, updating, or tiered context management (Park et al., 2023; Zhong et al., 2024; Packer et al., 2023). Mem0, A-Mem, and HippoRAG develop extracted memories, linked records, and graph-based retrieval (Chhikara et al., 2025; Xu et al., 2025; Gutiérrez et al., 2024), while LoCoMo and LongMemEval evaluate long-horizon recall (Maharana et al., 2024; Wu et al., 2025). Our comparison concerns how retained information becomes decision evidence: CSM supplies recursive summaries, while Graphiti retrieves event-derived relations with temporal validity and provenance (Zep, 2026). ProEviMem instead represents recurring conditional patterns, assesses their histor-

Method	AS		AL		RA		Mean	
	Prec.	F1	Prec.	F1	Prec.	F1	Prec.	F1
w/o Replay Gate	31.1	42.7	43.6	54.4	33.8	48.1	36.2	48.4
w/o Daily Validation	34.4	47.5	45.9	56.7	36.1	51.5	38.8	51.9
Before suppression	—	—	—	—	36.7	53.6	—	—
ProEviMem	35.8	48.8	48.6	58.6	40.1	55.4	41.5	54.3

Table 4: Evidence-check ablations and controller diagnostic: Precision and Intervention F1 (% , both $\uparrow$). Validation rows remove the named check; Before suppression scores recorded proposals, not a controller-free rerun. Mean equally averages three scopes. Appendix H adds intervention rates and recall.

ical support, and selects their current applicability. Section 5.2 compares these interfaces; Section 5.4 tests the evidence checks.

Proactive decision and memory. ProactiveAgent studies whether an agent should propose a task or remain silent and uses annotated event data to elicit this capability (Lu et al., 2025). PASK combines streaming demand detection with hybrid long-term memory, MemCog makes structured memory exploration part of conversational reasoning, and Remember When It Matters uses a separate memory agent to inject selected reminders into long-horizon action agents (Xie et al., 2026; Li et al., 2026; Wu et al., 2026). ProEviMem complements these directions by separating three evidence questions: which source facts support an abstraction, whether a proposed pattern has historical support, and when it applies to an incoming event.

Evaluation of proactive service. Existing benchmarks examine proactive task proposals, latent-need detection, or proactive memory triggering (Lu et al., 2025; Xie et al., 2026; Li et al., 2026). ProactiveStream evaluates repeated INTER- VENE/ABSTAIN decisions over causally replayed event streams, with entity isolation and episode-aware reference opportunities. Research on notification timing further shows that interruption cost depends on task phase and receptivity (Adamczyk and Bailey, 2004; Iqbal and Bailey, 2007, 2008; Mehrotra et al., 2016). This motivates evaluating opportunity recovery together with unnecessary intervention.

7 Conclusion

We presented ProEviMem, which turns longitudinal observations into source-checked daily trajectories and replay-assessed temporal patterns for proactive service decisions. On ProactiveStream, it improves Intervention F1 and Balanced Accuracy

over four baselines across three scopes; ablations support the hierarchy and its evidence checks. The resulting design treats history as traceable evidence with temporal applicability, while leaving present service value and opportunity novelty to the decision process.

Limitations

ProactiveStream covers three low-risk service scopes across 17 isolated streams. Reference labels denote team-reviewed opportunities under the specified episode policy, not observed recipient preferences or optimal actions. The additional review uses consensus labels from a separate three-reviewer team blinded to model outputs on 150 contiguous targets per scope, each from one stream; pre-discussion inter-annotator agreement is not quantified.

We use protocol-fixed evaluation without an independent development/test split. For the main comparisons, event inventories, reference labels, prompts, memory boundaries, and validation settings were fixed before the reported runs. This supports within-protocol comparisons, not generalization to held-out entities or domains. The paired 450-target review measures label sensitivity on three selected streams, not benchmark-wide ranking stability; the small AS F1 ordering changes with the label version (Appendix D). The paired episode diagnostic uses a separate model-assisted reference set and partially confirmed follow-up judgments; remaining semantic and boundary uncertainty limits conclusions beyond this descriptive CSM-ProEviMem comparison (Appendix D.1). The RA suppression analysis holds the recorded decision sequence fixed and does not estimate how removing the controller would change later agent states. Terra/Sol jointly replace decision and memory-construction LLMs, assessing pipeline sensitivity rather than isolated decision-model effects or cross-family transfer.

Daily validation and historical replay check

source support and recurrence in the observed history. These checks neither establish evidence completeness nor guarantee that patterns remain valid as behavior changes. Across scopes, offline evaluation measures intervention decisions, not verified service delivery, recipient satisfaction, realized interruption burden, long-term behavioral effects, or deployment safety. These outcomes require prospective evaluation with recipients; high-risk health and safety interventions remain outside the present scope.

Ethical Considerations

Our evaluation is limited to optional activity support, gentle check-ins, and low-risk repository assistance. It does not establish a basis for medical or high-risk safety decisions. Inference is isolated by stream, and reference labels, episode annotations, and scoring metadata are excluded from model inputs.

Longitudinal events and derived memories can reveal sensitive routines even when identifiers are pseudonymized. Event content, prediction traces, and inferred trajectories require access and retention protections; external-model use should minimize transmitted information and respect source-data permissions.

Deployment should require informed opt-in and allow inspection, correction, deletion, retention control, and disabling proactive assistance. An intervention prediction should be treated as a service proposal, not permission for consequential action. These deployment safeguards require separate implementation and evaluation.

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A Pipeline-Level Model Sensitivity

Table 5 provides the complete model-sensitivity comparison summarized in Section 5.2. The decision LLM and the LLMs constructing daily trajectories and temporal patterns are replaced together. The benchmark, memory architecture, role-specific prompts, decoding configuration, validation thresholds, and controller remain fixed; the resulting memory content may change. ProEviMem shows increasing Intervention F1 and decreasing False Intervention Rate from Luna to Terra to Sol in every scope. This is a pipeline-level comparison, not an isolated decision-model effect. All hierarchy and validation ablations retain Luna.

Scope	Model	Intervention F1 ↑	False Intervention Rate ↓
AS	Luna	48.8	46.1
	Terra	54.1	39.4
	Sol	58.6	34.4
AL	Luna	58.6	10.3
	Terra	64.4	9.1
	Sol	69.3	7.5
RA	Luna	55.4	18.1
	Terra	60.3	14.9
	Sol	66.4	11.8
Mean	Luna	54.3	24.8
	Terra	59.6	21.1
	Sol	64.8	17.9

Table 5: ProEviMem model sensitivity (%). Decision and construction LLMs change together. Bold marks the best value per scope; Mean follows Table 2.

B Memory-Construction Audit and Online Availability

Complementing the evidence-check ablations in Section 5.4, we audit existing memory-construction artifacts and maintenance logs from Activity Support and Ambient Living, covering nine participants and four homes. Table 6 distinguishes validation and replay operations from unique retained artifacts. The audit checks source support, entity consistency, and replay consistency rather than intervention outcomes.

Retained evidence. The daily store contains 423 trajectories—367 from Activity Support and 56 from Ambient Living—comprising 1,279 retained segments. All retained segments pass the configured source-support and entity-consistency checks, with no cross-entity evidence. Separately, construction logs record 311 rejected daily validation

checks. These observations characterize filtering and the retained evidence, rather than an error-free construction LLM.

The final temporal-pattern store contains 54 source-linked proposals whose recorded replay results are consistent with offline recomputation: 42 are Active and 12 remain Candidate. Under ProEviMem, Candidates remain excluded from decision evidence. Across weekly maintenance updates, 118 of 140 replay attempts pass the activation gate, while 22 do not. These operation counts are distinct from the 42 unique Active patterns in the final store.

Online availability. Daily Trajectory Memory first becomes available for all 13 entities at a median elapsed time of 0.80 days. An Active temporal pattern first becomes available for 11 entities at a median of 6.79 days, whereas relevant temporal-pattern evidence is first matched for eight entities at a median of 10.46 days. Each median is calculated over entities reaching the corresponding stage. Pattern activation and current relevance are therefore distinct: an admitted pattern does not automatically enter every decision context.

These observations show that daily and temporal-pattern evidence can become available within the evaluated streams, leaving subsequent events for evidence-informed decisions. The update order in Section 3.3 preserves the causal boundary: evidence used to judge an event is constructed from previously committed events. Together with Section 5.4, this audit addresses RQ3 by separating evidence integrity and availability from downstream service-opportunity detection performance.

View	Counting unit	Audit result
Daily validation log	validation attempt	311 rejected before storage
Retained daily store	unique evidence-linked segment	1,279 checked; all source-supported; 0 cross-entity
Weekly replay log	replay attempt	140 attempts; 118 pass; 22 not activated at that update
Retained pattern store	unique temporal-pattern proposal	54 checked; 42 Active; 12 Candidate; 0 inconsistent re-plays

Table 6: Memory-construction audit for Activity Support and Ambient Living (nine participants and four homes). Log rows count maintenance operations; store rows count unique retained artifacts.

C Illustrative Decisions and Error Cases

Table 7 traces one recorded AS decision from the 450-target diagnostic run, selected because its source events, daily artifacts, contemporaneous weekly proposal, and matched-pattern telemetry can be linked. It uses the proposal available at decision time, not a later revision from the final memory snapshot. The example illustrates evidence flow rather than a memory-attributable true positive or superiority over a baseline.

The current step counts provide an observable reason not to equate the earlier STILL pattern with a present need for activity support. This is our interpretation of the fields, not a recorded explanation from the decision LLM. The logged ABSTAIN originates from that LLM rather than duplicate suppression, preserving the distinction between pattern applicability and intervention judgment.

The separate frozen 16-case inventory includes two false positives and two false negatives from Event-only logs. In Activity Support, an activity suggestion is issued despite substantial movement during the preceding hour, while another low-activity event with a positive reference label receives ABSTAIN. In Ambient Living, ordinary bedroom–bathroom movement receives a check-in, whereas a reference-positive quiet period spanning two consecutive segments is missed.

These errors illustrate the distinction between an observable condition and an actionable service opportunity. Current inactivity or room transitions alone may be insufficient to justify assistance, but conservative interpretation can also miss reviewed opportunities. Historical evidence can contextualize such observations without determining the intervention decision by itself. Repeated windows additionally require assessing whether evidence or service needs have changed, complementing the controller trade-off examined in Section 5.4.

This qualitative analysis addresses the failure-analysis part of RQ4 and illustrates why ProEviMem separates memory evidence, present service

value, and prior opportunity coverage.

D Independent Human Review Across Three Scopes

We conduct a sampled label review within the main benchmark, selecting 150 consecutive target events per scope (450 in total). A separate three-reviewer team independently annotates these segments without access to either ProEviMem or CSM outputs, and discusses its judgments to obtain consensus references. Each segment belongs to exactly one participant, home, or project repository; no segment mixes entities. A fixed SHA256-based rule first ranks eligible streams and then eligible contiguous starts using public metadata only, without reference labels or predictions, and requires at least seven days of preceding history. The selected streams are AS participant_35, AL home_0004, and the RA repository callstack/react-native-pager-view. Their target periods are December 16, 2015–January 18, 2016; June 23–28, 2011; and June 21–November 4, 2025, respectively. All 37, 203, and 152 preceding events are retained and processed chronologically by the corresponding ProEviMem pipeline, but excluded from scoring. Thus, 842 public events supply causal context for 450 scored targets; each target can use only its current observation and strictly prior same-stream history.

The reviewers inspect each target with its preceding same-stream history, rather than judging isolated events. The confirmed review yields 27/123, 15/135, and 34/116 INTERVENE/ABSTAIN labels for AS, AL, and RA, respectively. Agreement with the earlier diagnostic references is 88.7% (133/150), 91.3% (137/150), and 85.3% (128/150) for AS, AL, and RA, respectively. In AS and AL, 13 and 13 labels change from INTERVENE to ABSTAIN, compared with 4 and 0 changes in the reverse direction; RA has 6 removals and 16 additions. Most judgments therefore agree, while service-opportunity boundaries remain consequen-

Stage and recorded time	Source-linked observation or operation
Historical events Dec. 15, 17–19, 21	Participant 35 has recorded STILL observations in slots 2–4 across five cited days. For example, Dec. 15 at 16:32:38 records slot 2, work, STILL, and 586 steps in the preceding hour (event 5fd72a80. . .). STILL describes a snapshot, not continuous inactivity.
Daily construction Dec. 16–22	Each cited day becomes a source-linked daily trajectory at the next daily rollover. The Dec. 15 trajectory retains the example’s event reference and observed fields; its recorded creation time is Dec. 16 at 12:02:49.
Weekly proposal Dec. 22	At the weekly update on Dec. 22, the proposal cites the five completed trajectories dated Dec. 15, 17–19, and 21: recognized activity repeatedly equals STILL in slots 2–4, with applicability window 16:00–22:00. The maintenance record at 12:02:27 reports one proposed and one activated pattern.
Current event Dec. 24, 16:31:26	Event c956aa51. . . : slot 2, available, not in transit, store location, activity UNKNOWN, 256 steps in the preceding 30 minutes and 778 in the preceding hour. The context contains 12 prior events, two daily trajectories, and the matched pattern 93463c34. . .
Decision, then commit	Both model proposal and final output are ABSTAIN; controller action is ALLOW with no suppression. Telemetry records that the current event is committed after context construction. A matched recurring observation therefore need not produce an intervention.

Table 7: A source-to-decision timeline for one AS participant. Dates are in December 2015; times use the stored UTC timestamps, not inferred local activity times. Text condenses source events and construction/decision logs.

tial. These counts characterize differences between label versions, not pre-discussion inter-annotator agreement or the semantic cause of each disagreement. These annotations serve as references for the additional evaluation; the main benchmark labels and result tables remain unchanged. For model comparison, we use the recorded ProEviMem and CSM decisions from the selected-segment diagnostic runs, rather than a subset of the main-run output files. For each method, exactly the same target decisions are scored against both label versions; changing the reference labels does not rerun inference or alter the outputs. Like the main experiment, this diagnostic uses gpt-5.6-luna through the official OpenAI API, with temperature zero and thinking disabled. Decision-output limits are 256 tokens for AS/AL and 512 for RA. Both methods retain the 72-hour/100-event bound and episode control; ProEviMem additionally retains daily validation and replay gating.

Table 9 reports ProEviMem under the confirmed event-level references, and Table 8 adds CSM under both label versions. ProEviMem identifies 24/27 AS, 15/15 AL, and 18/34 RA opportunities. Its F1 remains above CSM in AL and RA under both versions; the small AS difference changes direction. Thus, the paired audit reveals both agreement and label sensitivity, rather than establishing invariant method rankings. For example, AL ProEviMem F1 changes from 68.7% to 55.6% when only the references change, while CSM changes from 46.4% to 37.2%. These are within-segment comparisons, not changes from the

full-scope main results: the sampled events and diagnostic runs differ from the main evaluation. Across scopes, immediate and scheduled service decisions share the INTERVENE label. The following paired episode analysis uses a separate, provisional reference set grounded in these reviewed events and their causal histories; event-label confirmation does not by itself confirm episode boundaries or message matches.

Scope	Method	Earlier F1	Reviewed F1
AS	CSM	44.4	41.0
	ProEviMem	42.6	42.5
AL	CSM	46.4	37.2
	ProEviMem	68.7	55.6
RA	CSM	49.4	46.5
	ProEviMem	57.6	47.4

Table 8: Paired reference-label sensitivity on 150 targets per scope. F1 is in percent. Within each method, the recorded diagnostic-run decisions are identical in both columns; only the reference version changes. Earlier references are original labels for AS/AL; RA uses 122 original labels and 28 subsequently completed human labels, rather than an entirely pre-test-frozen comparator.

D.1 Paired Episode-Level Behavioral Diagnosis

On the same 450 targets and 392 preceding events, we compare recorded CSM and ProEviMem decisions using a common set of 67 proposed episodes: 26 in AS, 10 in AL, and 31 in RA. The analysis separates first effective assistance, useful follow-up, and repetition, without additional inference or

Quantity	AS	AL	RA
Scored decisions	150	150	150
Preceding events (not scored)	37	203	152
Reference interventions	27	15	34
Reference abstentions	123	135	116
ProEviMem interventions	86	39	42
TP	24	15	18
FP	62	24	24
FN	3	0	16
TN	61	111	92
Intervention Rate	57.3%	26.0%	28.0%
Precision	27.9%	38.5%	42.9%
Opportunity Recall	88.9%	100.0%	52.9%
Intervention F1	42.5%	55.6%	47.4%
False Intervention Rate	50.4%	17.8%	20.7%
Balanced Accuracy	69.2%	91.1%	66.1%

Table 9: Recorded ProEviMem decisions evaluated against consensus labels from a team blinded to model outputs, on 450 targets with 150 contiguous decisions per scope. Each segment and its prefix are entity-isolated. Counts use final intervention decisions after episode control, including immediate and scheduled outputs. Prefix events supply context only; main benchmark results remain unchanged.

changes to event labels. Each episode specifies a service goal and object, causal evidence, onset, valid observed event positions, and any explicit deadline or closure evidence; it is not a daily trajectory or temporal-pattern record. The reference set and semantic matches are model-assisted proposals. Six AL follow-up classifications have human confirmation; the remaining proposals are not treated as independently confirmed annotations. We therefore report this version as a provisional, post-hoc paired diagnostic, separate from the blinded event-label review. Validity at listed observations does not imply continuous validity between sparse events.

Reference and model histories. Fixed event labels use reference-opportunity novelty (Equation 9); episode coverage instead follows the evaluated method’s actual preceding interventions. Each method initializes coverage from its own prefix outputs, without counting prefix interventions as target outcomes or borrowing the other method’s coverage history. A target intervention must match an episode’s service goal and object at a listed valid observation. For timeliness, intended service time is the decision time plus the specified delay, with zero delay for immediate intervention. If an explicit reference deadline exists, an intended service time beyond it receives no coverage credit. The first timely match covers the episode. For subsequent

matches, we compare the message and available evidence with that method’s preceding assistance for the same episode. A follow-up is valuable when new causal evidence changes the actionable service or next step and the message addresses that change; rephrasing the same advice or continued status alone is redundant. Cases requiring unobserved delivery, recipient response, or task completion are left uncertain. Valuable follow-ups do not create extra episodes or increase coverage. Each output credits at most one episode: among compatible references, the first uncovered one in the fixed reference order takes precedence; otherwise the first already covered one is used. Unmatched or late outputs remain in the intervention denominator. The original event labels and main result tables are unchanged.

Metrics. Let I be all target interventions, F first valid interventions, V valuable follow-ups, and R redundant follow-ups. Episode Coverage is covered new episodes divided by all new episodes. Supported Intervention Precision (SIP) is $(F + V)/I$, extending the first-only Episode Precision F/I without giving extra coverage credit. Redundant Intervention Rate (RIR) is R/I . Uncertain follow-ups (U), unmatched outputs, late outputs, and unresolved matches remain in I , but not in $F + V$ or R . These categories describe compatibility with the proposed references, not verified recipient benefit. Table 10 reports coverage and rates; Table 11 retains the count decomposition.

Observation boundaries. The main diagnostic treats the first AL opportunity as target-onset; a plausible continuation from the prefix changes both methods’ denominator to nine, with CSM covering 7/9 and ProEviMem 9/9. Coverage describes the observed segment, not an opportunity’s full future lifetime. RA has six late CSM outputs and four late ProEviMem outputs; scheduled decisions are checked against explicit deadlines, not verified delivery. Three CSM RA follow-ups have uncertain value: accepting all three would raise its SIP from 33.8% to 38.5%, a sensitivity scenario rather than an uncertainty interval. Other unmatched and unresolved outputs are not exhaustively adjudicated for alternative service value.

Behavioral comparison. AS coverage is identical, while ProEviMem uses fewer interventions. In AL, ProEviMem covers two more proposed opportunities and provides more valuable follow-ups,

Scope	Method	Covered / New	Coverage $\uparrow$	SIP $\uparrow$	RIR $\downarrow$
AS	CSM	23/26	88.5	26.7	1.1
	ProEviMem	23/26	88.5	27.9	1.2
AL	CSM	8/10	80.0	39.3	32.1
	ProEviMem	10/10	100.0	41.0	43.6
RA	CSM	18/31	58.1	33.8	4.6
	ProEviMem	14/31	45.2	40.5	9.5

Table 10: Provisional paired episode diagnostic on 150 targets per scope, using the same proposed references and each method’s causal history. Rates are percentages. SIP credits first valid assistance and valuable follow-up; RIR counts only redundant follow-up. Bold marks the better unrounded directional value within each scope (or both for ties). These are descriptive scores under partially confirmed annotations, not a separate confirmatory evaluation.

Scope	Method	<i>I</i>	<i>F</i>	<i>V</i>	<i>R</i>	<i>U</i>	Unmatched	Late	Unresolved
AS	CSM	90	23	1	1	0	64	0	1
	ProEviMem	86	23	1	1	0	60	0	1
AL	CSM	28	8	3	9	0	8	0	0
	ProEviMem	39	10	6	17	0	6	0	0
RA	CSM	65	18	4	3	3	30	6	1
	ProEviMem	42	14	3	4	0	16	4	1

Table 11: Intervention counts underlying Table 10. *I* equals the sum of all seven outcome categories. *U* denotes uncertain follow-up value; Unresolved denotes a pending semantic or condition match. Both remain separate from redundant or supported assistance. A valuable follow-up does not increase the episode count.

but also has a higher redundant-intervention rate. In RA, CSM covers more opportunities, whereas ProEviMem has higher SIP with fewer interventions. The low supported shares partly reflect outputs unmatched to this finite reference inventory; they do not establish that every remaining message is useless. These trade-offs distinguish coverage, supported follow-up, and repetition rather than imply uniform superiority or isolate the controller’s causal effect.

E Memory Implementation Details

This appendix expands the evidence lifecycle in Section 3; the core acceptance conditions and causal interfaces remain in the main paper.

E.1 Daily Buffer and Checkable Fields

Daily construction uses an independent buffer of committed events for the open local day, so Event Memory truncation does not discard construction evidence. The buffer is cleared after that day is processed. Each daily segment records a local interval, typed observed, context-before, and outcome fields, and supporting event identifiers. These activity segments are distinct from service episodes used for intervention control. Scalar values are canonicalized; field paths must agree exactly or

through a schema-preserving suffix. The validator also checks entity, interval, and citation consistency. Unsupported facts and unresolved or cross-entity citations are rejected. Free-text descriptions do not substitute for typed evidence in validation or replay. Accepted segments form daily trajectories retained in the current construction cycle, even after their source events leave Event Memory.

E.2 Construction Clock and Replay Boundaries

A weekly cycle is a non-overlapping, time-ordered block of seven completed daily-trajectory records for one entity, not a fixed calendar week. Dates without a daily record are not filled in, so a cycle may span more than seven calendar days. Empty or unobservable records count toward cycle completion but provide no fulfilled or violated replay evidence. After consolidation, the online daily-memory view starts a new cycle, while daily records remain available for audit.

A proposal specifies trigger context *g*, expected pattern *p*, a local-time window, optional weekday restrictions, and supporting daily identifiers. It describes an expectation, not a recommendation to intervene. Replay excludes days outside applicability conditions and censors missing observations.

An observable day with a matching trigger is fulfilled if p occurs at or after the trigger within the specified window; otherwise it is violated. Each date contributes at most once. Censored days do not enter the confidence denominator in Equation 6. Confidence is zero without eligible days; lift is zero when the base rate is zero. The base rate uses observable days under the same window and weekday restrictions, without conditioning on g .

E.3 Pattern Lifecycle

Lifecycle status controls retrieval eligibility, separately from current occurrence and service-episode states. After admission, fulfilled and violated observations update stored confidence through a smoothed support ratio combining replay counts and subsequent outcomes with one initial success and one initial failure. Censored observations supply neither success nor failure evidence. Two consecutive violations move an Active pattern to *weakening*, which remains retrievable at lower priority; four move it to *dormant*, which is excluded. A fulfilled observation restores a weakening pattern to Active when updated confidence is at least 0.5. A dormant pattern returns only to Candidate and requires replay readmission. Explicitly retired patterns and all Candidates are excluded from ProEviMem retrieval.

E.4 Current Occurrence Matching

A deterministic matcher checks trigger context, weekday restrictions, and expected windows using a 30-minute lead and grace interval. Triggers may be current or previously recorded causal occurrences. Each occurrence has one of five states: not-started, pending, fulfilled, deviated, or censored. Fulfilled and censored occurrences are excluded. The remainder are ranked by occurrence state (deviated, pending, then not-started), lifecycle priority, stored confidence, and temporal distance; at most three patterns enter R_t . All matching and state assessment use only the current event and causally available observations. Daily Trajectory Memory supplies the current cycle’s completed daily trajectories rather than a similarity-ranked subset.

E.5 Episode Identity and Causal Execution

The controller forms an intervention key from scope-specific episode state, proposed service action and target, and sorted pattern identifiers present in the decision context. When no pattern identifiers are present, that part of the key is an

empty list; identifiers are not inferred from plain summaries or graph facts. State includes availability/activity context for AS, activity/room context for AL, and repository activity/target identifiers for RA. AL additionally distinguishes onset, confirmed, and sustained phases at one, two, and three or more consecutive matching observations, respectively. When an explicit episode key is unavailable, scope-specific time or target fields identify the opportunity, and unchanged evidence is additionally required for duplicate suppression. Changed episode state, service, pattern identifiers, or local day can therefore produce a different intervention key; this permits reconsideration but does not itself request intervention. ProactiveAgent, Mem0, CSM, Graphiti, Event-only, Event+Daily, and ProEviMem share this policy, not identical keys: different available evidence can change the identity assigned to a proposal.

The causal execution order is:

1. Prepare prior memory; any due maintenance uses only previously committed events.
2. Stage e_t without committing it, then select daily and temporal evidence.
3. Obtain the decision LLM’s proposal from the structured context.
4. Apply repeated-opportunity suppression and finalize the output.
5. Commit e_t and update coverage state; maintenance incorporating e_t affects only later decisions.

Reference labels are joined only for offline scoring.

F Public Events and Reference-Label Protocol

F.1 Scopes and Public Events

Activity Support. This scope derives from HeartSteps mobile activity contexts (Klasnja et al., 2019). We retain 1,456 meta-events from nine complete participant streams, covering 367 observed participant-days. Complete streams preserve the chronology required for daily and longer-term memory construction without sampling within a participant. Public fields include decision-time availability, recognized activity, recent step summaries, location category, weather, and local decision slot. Randomized treatment assignments

and future behavioral outcomes are excluded. Reference labels concern opportunities for optional activity support, rather than reproduction of the original intervention assignment or estimation of health effects.

Ambient Living. This scope derives from CASAS ambient-sensor traces (Cook et al., 2013) and contains four homes with 14 consecutive days each. Within every local day, ten-minute pre-processing blocks are merged into 28 regular 50-minute segments and one 40-minute tail, producing 406 events per home. Every source block is assigned to exactly one segment, preserving the complete two-week chronology. A segment becomes available only after its interval closes and contains within-interval motion counts, room sequences, door actions, and compact numeric summaries. It neither summarizes earlier segments nor exposes a personal baseline, deviation score, or candidate judgment; historical evidence must therefore be recovered by the evaluated agent. The service space is restricted to gentle, low-risk check-ins and excludes disease, fall, travel, danger, and unsupported preference inference.

Repository Assistance. This scope organizes 1,459 repository activity records into four isolated streams, with 174 reference interventions (Table 1). Each event represents a timestamped repository activity, such as a commit push, issue update, pull-request action, or review. We preserve GitHub’s native event granularity, with each event corresponding to one intervention decision. Within each repository, records are ordered by event time, with event ID used to break ties deterministically. The source inventory includes public GH Archive records (Grigorik, 2026). Each repository independently maintains Event Memory, Daily Trajectory Memory, and Temporal Pattern Memory, together with its intervention history and episode state. Observations, extracted evidence, and coverage state are not transferred between repositories, even when contributors participate in multiple projects. The memory owner and evaluation unit is therefore the repository stream rather than an individual contributor. Services include low-risk triage, review, and maintenance assistance.

Public-input contract. Reference-positive and reference-negative events follow the same scope-specific public schema. Public inputs contain event identifiers, stream identifiers, timestamps, and con-

temporaneously observable fields, but no reference label, candidate flag, episode annotation, trajectory summary, memory-level marker, or scoring mask. The inventory includes early events observed before Daily Trajectory Memory or Temporal Pattern Memory becomes available.

F.2 Reference Labels and Causal Replay

Service-opportunity semantics. Annotators assess the current event e_t and the complete strictly prior public history $\mathcal{H}_{<t} = (e_1, \dots, e_{t-1})$ of the same entity, independently of the memory representation retained by a tested method. Let $A(e_t)$ indicate that the current event itself establishes a concrete, timely, feasible, and low-risk service opportunity, and let $B(e_t, \mathcal{H}_{<t})$ indicate that prior evidence makes such an opportunity visible at the current event. Let N_t^{ref} indicate opportunity novelty under the reference episode protocol below. The reviewed label follows Equation 9. These predicates specify annotation semantics, rather than learned classifiers or additional decision gates. Historical deviation is neither necessary nor sufficient: an event may be actionable without a deviation, while an unusual event may still lack a useful or safe service action. Within a semantic episode, the first reliable and actionable window is labeled INTERVENE. Subsequent covered windows are labeled ABSTAIN unless the evidence strengthens, the feasible service changes, or the episode reopens. Reference episode coverage, and hence N_t^{ref} , is determined from the reviewed opportunity sequence up to the current event rather than from a method’s predicted interventions. The labels consequently represent reviewed service opportunities, not recipient acceptance, treatment effects, or an optimal deployment policy.

Annotation protocol. An LLM-assisted screening stage identifies potential service opportunities and expands the review queue. Three team members then independently assign initial labels without access to the evaluated systems’ decisions, followed by team cross-review and adjudication. Every public event enters final review, including events not selected during screening. The screening output can prioritize inspection but cannot determine a reference label automatically. We use protocol-fixed evaluation rather than claim an independent development/test split. The Ambient Living aggregation rule is specified before its event rows are constructed, and each evaluation version

jointly records its event inventory, reference-label map, intervention prompt, Event Memory boundary, and validation settings.

Causal replay. Every participant, home, or repository stream is replayed independently from empty method state and in timestamp order. At event t , a method may access the current event e_t , its own strictly prior same-stream memory, and any preceding outputs retained by its architecture. It may not access another stream, a future event, an evaluator-private label, or a summary computed from the complete sequence. The intervention decision is finalized before e_t is committed to memory. Daily and weekly maintenance involving e_t can therefore affect only later decisions. Reference labels are joined after inference for scoring and never enter memory construction or retrieval. Annotation artifacts, reference episode states, and scoring metadata remain organizer-private.

G Additional Experimental Configuration

External baselines. *ProactiveAgent* adapts ProactiveAgent’s released evaluation prompt and proposal/null output contract (Lu et al., 2025; THUNLP, 2026), without its fine-tuned checkpoint. It retains causal event–decision history, capped at 1,000 events including the current event per request, without external structured memory. *Mem0* uses the open-source mem0ai==2.0.12 extraction, consolidation, and retrieval pipeline (Chhikara et al., 2025; Mem0, 2026), local all-MiniLM-L6-v2 embeddings, top-8 same-entity retrieval, and a 0.1 search threshold. The baselines retain their method-specific memory and prompt interfaces while sharing the task semantics and final Episode Controller. This comparison does not impose equal context or token budgets.

Cross-week summary and temporal-graph baselines. CSM is an in-house plain-summary control, not a reimplement of an external summary agent. It summarizes each completed local day and, after seven completed daily records, combines that block with the previous cross-week summary. This recursive update retains information from earlier weeks rather than replacing it with the latest week alone. The decision context contains the current event, bounded Event Memory, completed daily summaries in the current block, and the latest cross-week summary. The adapter caps

daily and cross-week summaries at 2,000 and 6,000 characters, respectively. It applies output-format and length checks, but no ProEviMem typed-fact validation, replay thresholds, Candidate/Active lifecycle, or deterministic temporal-rule matching.

Graphiti uses the open-source temporal-graph framework (Zep, 2026) to ingest committed public events into a stream-isolated graph. The adapter uses Graphiti 0.30.2, Neo4j, and local 384-dimensional all-MiniLM-L6-v2 embeddings. The current public event is the retrieval query; native hybrid edge search with reciprocal-rank fusion returns at most ten relation facts, including validity fields and source-record identifiers. Retrieved identifiers are not expanded into neighboring source events or daily trajectories. Graphiti’s ingestion episodes denote source records, not service-opportunity episodes. Both baselines retain ProEviMem’s 72-hour/100-event bound for the event context and reuse the task policy, output contract, and Episode Controller with memory-specific guidance. Neither treats its summaries or graph facts as replay-assessed ProEviMem patterns. Their comparison isolates alternative memory interfaces, not identical prompt text or equal end-to-end computation.

Memory variants and retrieval control. *Event-only* receives the current event and bounded Event Memory (Equation 4). *Event+Daily* additionally receives fact-validated daily trajectories; ProEviMem also receives eligible temporal patterns (Equation 7). These three variants share the intervention prompt, output schema, and Episode Controller, differing in available memory evidence. For Activity Support and Ambient Living, *Flat Retrieval Memory* adds to Event-only the three most similar older same-entity events outside its window. It uses fixed dense embeddings and cosine similarity, without daily consolidation, replay, neighbor expansion, or LLM reranking.

Mechanism ablations and controller diagnostic. Section 5.4 defines two evidence-check ablations and a suppression diagnostic; Table 13 reports their intervention rates, precision, recall, and F1. Each validation ablation removes only its named check. The controller diagnostic compares recorded proposals with finalized outputs on the same RA sequence, without recomputing later states or decisions.

Model roles and implementation. In the primary setting, decision and memory-construction calls use gpt-5.6-luna through the official OpenAI API, with temperature zero and thinking disabled. Decision generation and memory construction are separate call roles, not a single shared output budget. Decision limits are interface-specific (256 or 512 tokens); construction-call limits are configured separately and recorded by role in the run configuration. Event Memory contains at most 99 prior events within 72 hours, plus the separately supplied current event. Daily construction follows local-day closure; weekly construction and replay follow seven completed daily records as defined in Section 3. All memory and validation records are entity-scoped.

We also test gpt-5.6-terra and gpt-5.6-sol. Both decision and construction LLMs change together; Appendix A provides the fixed settings and complete results. Input manifests, prompts, schemas, thresholds, and role-specific configurations are versioned in the experiment ledger.

Scoring. Section 5.1 defines the scoring policy and metrics. Intervention Rate measures decision frequency, not perceived interruption burden.

G.1 Intervention-Rate Interpretation

Intervention Rate is a behavioral operating point, not a standalone performance target. Writing ρ for reference-positive prevalence, the exact identity $IR = \rho \text{ Recall} + (1 - \rho) \text{ FIR}$ separates opportunity recovery from false intervention, with all rates expressed as fractions. The prevalences are 25.1%, 11.7%, and 11.9% in AS, AL, and RA. However, prevalence alone cannot explain the variation: ProactiveAgent intervenes at 9.2% of AL events but 53.1% of RA events despite their similar prevalences. Their respective recall and false-intervention rates reveal different coverage-selectivity operating points under the scope-specific evidence and service semantics. This descriptive decomposition does not isolate an intrinsic LLM proactivity trait from memory, prompt, and controller effects.

H Extended Ablation and Controller Results

Tables 12 and 13 expand main-paper Tables 3 and 4 with net TP increments, intervention rates, and recall. Net TP changes do not identify individual causal contributions. Intervention rates and recall

are retained to expose coverage-selectivity trade-offs, including the RA-only Episode Controller comparison.

Scope	Method	Intervention F1 $\uparrow$	FIR $\downarrow$	Daily Δ TP	Pattern Δ TP
AS	Event-only	38.3	51.3	–	–
	Event+Daily	38.9	51.3	+4	–
	ProEviMem	48.8	46.1	–	+57
AL	Event-only	40.0	12.4	–	–
	Event+Daily	45.1	12.4	+15	–
	ProEviMem	58.6	10.3	–	+33
RA	Event-only	28.7	18.9	–	–
	Event+Daily	48.5	19.1	+64	–
	ProEviMem	55.4	18.1	–	+22
Mean	Event-only	35.7	27.5	–	–
	Event+Daily	44.2	27.6	–	–
	ProEviMem	54.3	24.8	–	–

Table 12: Memory-hierarchy ablation. Intervention F1 and False Intervention Rate (FIR) are percentages; Δ TP is a net count. Daily Δ TP compares Event+Daily with Event-only; Pattern Δ TP compares ProEviMem with Event+Daily. The increments are not individual-event attributions. Bold marks higher F1 and lower FIR. Mean averages displayed scope metrics; count increments are not averaged.

Scope	Method	Intervention Rate	Precision $\uparrow$	Opportunity Recall $\uparrow$	Intervention F1 $\uparrow$
AS	w/o Replay Gate	55.0	31.1	68.2	42.7
	w/o Daily Validation	55.8	34.4	76.7	47.5
	ProEviMem	53.8	35.8	76.7	48.8
AL	w/o Replay Gate	19.3	43.6	72.1	54.4
	w/o Daily Validation	18.9	45.9	74.2	56.7
	ProEviMem	17.7	48.6	73.7	58.6
RA	w/o Replay Gate	29.4	33.8	83.3	48.1
	w/o Daily Validation	29.6	36.1	89.7	51.5
	Before suppression	32.4	36.7	99.4	53.6
	ProEviMem	26.7	40.1	89.7	55.4
Mean	w/o Replay Gate	34.6	36.2	74.5	48.4
	w/o Daily Validation	34.8	38.8	80.2	51.9
	ProEviMem	32.7	41.5	80.0	54.3

Table 13: Evidence-check ablations and controller diagnostic (%). Validation rows remove the named check; Before suppression scores proposals from the recorded RA run without recomputing later states. ProEviMem is last. Bold marks the best directional score. Mean covers the three scopes; the RA-only diagnostic has no mean.